

Setify: An Online Repository to Catalog Concert Setlist Playlists

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Abstract

This study researched the benefits of an online catalog of music artists' concert setlists, opening and headlining acts, and related information about the songs in concerts. Prior to this capstone project, there was no online system that allowed concert setlists to be categorized into playlists that include the opening acts' and headliners' setlists and a way to listen to the songs and view the lyrics. This project was necessary to allow fans to feel connected to artists' tours, even if they are not able to attend the concert, and to help fans discover new artists through the inclusion of the opening acts' setlists. The methodology was to create an online repository and catalog service that includes all of these features. Important factors in creating this online repository and catalog service were personalization in creating and viewing the playlists, best practices of software design, providing a way to listen to the songs, and providing a way to view the lyrics of the songs. Using tools such as Xcode, Swift language, MongoDB database, Node.js Express, and Postman for testing, an iOS application called Setify was created.

Introduction and Thesis Statement

The subject that was researched was “An online catalog of music artists concert setlists, opening and headlining acts, and related information about the songs in concerts”. There was a need for a system like this because music streaming services, such as Spotify, while extremely popular, do not have accurate setlists from concerts. In 2017, music streaming services accounted for 54% of all audio consumption (Epstein). Also, in 2019, the size of the market for music streaming services was \$20.9 billion, which grows by about 17.8% each year (Musicians Institute).

Personalization is an important factor in music streaming services because it makes users more likely to recommend the music streaming service to other people (Barata and Coelho). The system that was created is an online system that is a catalog of concert setlists, including opening and headlining acts, and related information about the songs in the concerts.

The system allows the user to create their own playlists of concert setlists, which is an example of personalization. This helps the success of the system, since it encourages more people to use it. A system that allows users to listen to playlists of concert setlists is important because it allows people who are not able to attend the

concert to feel connected to other fans and to the artist (Charron). The system uses the best practices of software design by having an organized user interface (U.S. Department of Health and Human Services). The system is intuitive with descriptive buttons and it uses the principle of hierarchy in order to make important headings larger than the other text. Before this system, there was no music cataloging system that had all of the components of personalization, best practices of software design, organizing both headliners and openers setlists into one tour playlist, providing a way to view the lyrics, and providing a way to listen to the songs.

In order to solve this problem, the system that was created is a music concert setlist repository service. This allows the user to post their own setlist playlists, browse playlists that are posted, select setlists as favorites, have links to listen to the songs in music streaming services, and view the lyrics of the songs. The system supports users creating concert setlists, browsing, and searching Spotify's API. The user is able to create a playlist by entering the opening artist's name, the opening artist's setlist, the headliner's name, the headliner's setlist, the tour name, the setlist name, the date, and the city. These features of the system make it user friendly and functional.

Literature Review

Music Streaming Services are Popular and Important

Music streaming services have become extremely popular in recent years. It has become the new norm of listening to music. As of 2017, on-demand audio streaming accounts for 54% of all audio consumption, while physical albums account for only 21% of all audio consumption and digital albums/digital track sales account for only 24% of all audio consumption (Epstein). In 2019, the market size of music streaming services was 20.9 billion, and each year it grows by about 17.8%. Also, 80% of all revenue for the music industry was from music streaming (Musicians Institute).

Streaming music digitally is quick, easy, and efficient. Music streaming services, such as Spotify, provide users with thousands of songs on demand. Music streaming services are important because they allow users to listen to music from around the world (Way). This allows users to be able to listen to music from artists that are located in other countries or states without traveling.

With the advancement of technology and algorithms over the years, the music streaming services are becoming more advanced and useful to consumers and

stakeholders, such as the artists and record labels. This is because the music streaming services are producing valuable data and information about users, and they increase revenue for artists and the industry (Maasø). This will continue to increase the popularity of music streaming services because they will continue to become more useful to users and the stakeholders with the additions of features and the creation of more music streaming services. When a song is added to Spotify's "Today's Top Hits" playlist, it will increase the number of streams of that song by about 20 million streams. This generates about \$116,000-\$163,000 in revenue (Aguar and Waldfogel). This shows that playlists on music streaming services are beneficial to the artists.

Music streaming is also popular because it has features that allow the user to create their own experience and personalize their profiles to reflect their interests.

Personalization is an Important Factor of Music Streaming Services

Personalization is an important feature of music streaming services. Personalization is a feature of these music streaming services that increases user's probability of recommending the music streaming service to others (Barata and Coelho). This is because it improves their listening experience so they can listen to exactly what they want to listen to. An example of a personalization factor can be the ability to create

playlists on the music streaming services. This is important because it shows that users want the ability to personalize their listening and create playlists of music they want to listen to. It is more beneficial than listening to individual songs.

Personalization causes the user to enjoy the music streaming service more through the user being able to create playlists, see information about the songs, read the lyrics, and more. It is beneficial to have an online system that has many options of playlists for concerts. For example, if a person is going to attend a concert, it is likely that they will like the songs being played at the concert. A playlist of the concert setlist will give the person another option to listen to what they like.

Personalization is important because it will get the user to listen to the music on the online system more. Listening to music is proven to release oxytocin (Suttie). This is a chemical in the body that is proven to increase bonding and trust. Since music is proven to have a positive impact, it is important to have factors on music streaming services that get people to listen to music more, such as personalization of playlists. It is unlikely that people will want to listen to many songs that they do not like.

Therefore, personalization in creating playlists of songs they like will increase their likelihood of listening to music and having oxytocin released, causing a positive effect

on their body. It will also cause them to use the online system more since they have a positive experience.

Music Streaming Services are Important to the Users and the Artists

Music streaming services are beneficial to both the user and to the artists who have songs on the platforms. As the platforms become more and more popular, artists measure their success on the number of streams they have instead of the number of albums they sold. For artists, having their music on playlists is very helpful. It allows more people to hear their music. The created setlist cataloging service shows playlists of full tours. This includes the opener's setlist and the headliners setlist. In a tour, the opener is a less famous artist who performs before the headliner of the tour. By having playlists of concerts that include the opener's setlist as well as the headliners setlist, it allows the less popular opener of the tour to have their music streamed by more people, which are the many fans of the headliner of the tour. This is helpful to the opener because they can grow their fanbase. It is also beneficial to the listener because they are able to get familiar with the opener's songs before they see them in concert.

Music streaming services also act as a social network, since users can post the

playlists that they create. (Epstein). They can view others playlists and discover new music.

Touring for artists is very important. It is a way for them to connect with their fans and make money for themselves and their record labels (Lashua, et al). Touring is a way for artists and fans to be in the same place. They all have a common interest and want to have fun at the concert. Music streaming services are connected to tours and live concerts because people who attend the concert and people who do not attend the concert can benefit from playlists of the concert setlist online. People who are not able to attend concerts can still feel like they are a part of it by listening to the music that is played at the concerts through playlists of the setlists of the concert (Charron). Many people cannot afford to attend concerts because concert ticket prices are often expensive. Many times, the prices depend on the demand there is for that concert. The higher the demand, the higher the ticket prices (Volpano). Having an online system that allows the user to listen to the songs played at the concert will help them to feel connected to the artist and other fans. Technology has impacted the live music category in many ways. This is because “live music is streamed, shared, uploaded, downloaded, reviewed, watched, and re-watched online” (Jones and Bennett). People

stream music because they want to experience performances even if they cannot attend them live. Also, people who attend concerts like to show that they were there to others, since it shows they are true fans of the artists. These people are the ones who post the setlists to show it to others.

Tours are also important to music streaming because it was found that there was a positive correlation between a concert and streaming of that music for up to 10 weeks after the concert (Hogue). This shows that concerts help artists because it increases the amount that their music is streamed. It was also found that streaming has a positive effect on live music sales and attendance for artists on streaming services (Nguyen et al). This shows that touring positively affects music streaming, and music streaming positively affects touring.

Another reason that music streaming services are important to the users and the artists is that music streaming services give users the ability to become fans of new artists.

When music consumption is free, and the music streaming service does not require a paid subscription, users listen to more of a variety of music (Datta, et al). This allows users to listen to music they might not have paid for if it was not free. Although there are premium versions of some music streaming services, for example Spotify

Premium, Spotify also has a free version that allows users to listen to songs for free with a limit of 6 skips per hour. This encourages users to listen to a variety of music which benefits the artists.

Viewing Lyrics are Important to the User Experience

Seeing the lyrics of a song helps improve the experience of listening to it. The created online system also has an option to view the lyrics of the songs. This is because viewing the lyrics allows the listener to understand the meaning behind the song in a better way. For example, if the music sounds upbeat and happy, but the lyrics are sad, the listener will most likely perceive the song as happy. However, when they look at the lyrics, they can see what the song is about in a better way. It has been shown that while listening to songs, the emotions that are felt are stronger when the lyrics are combined with the music (Brattico, et al). By reading and understanding the lyrics, it will improve the listener's emotional response to the songs. This will improve the listening experience because the listener will be able to listen to the song in the way the artist meant for it to be listened to. This is helpful to both the artist and the listener. This is because it creates a better relationship between the listener and the artist, which helps create an aspect of social networking to the online system (Epstein). By

connecting with others through listening to their playlists, it makes the online system have a social network aspect. People are able to connect to each other through music. Also, having a way to view the lyrics is an important part of the system because it allows users to sing along to the songs without memorizing the lyrics before they sing along (Noor). This allows the user to learn the lyrics in a quicker way while they listen to the song.

Elements of An Effective Online System and Importance of a User-Friendly Interface

Creating an online system has many important factors in creating a user-friendly interface. Readability is an important factor, since it is defined as “The collection of all elements in textual material that affects level of understanding of the reader” (Ojha). It allows the user to understand the system and use it correctly. There are many features that improve a user’s experience on an online system (Garett, et al).

Organization is a factor that impacts readability. Organization is an important factor because if the system does not have an organized interface, the user will not be able to use it correctly. They might not be able to access parts of the system, and they will not get the benefit of those features that they cannot access due to a lack of organization.

This also shows that the system has to be able to be navigated easily by the user. It

should be intuitive, with descriptive buttons and headings. Utilizing colors and images also improves the user experience and makes the system more interesting. The system should be simple, reliable, have accurate information, have good security, be fast, interactive (posting and viewing other people's posts), and give the user the ability to customize their experience. These features of an online system are important because they will cause the user to use the system more and will improve the probability of them recommending it to others.

The U.S. Department of Health and Human Services also has guidelines on how to create a user-friendly website (U.S. Department of Health and Human Services).

These guidelines can be applied to all online systems. A guideline is to make the system have consistent elements, such as having consistent labels. This will help the user be able to learn how to use the system quicker. If they learn what a label means in one place on the system, they should be able to know what the same label means when it is used throughout the system. Hierarchy is also an important guideline. This means that the headings of a system should be larger to get the user's attention. It helps the user scan the system and be able to find what they are looking for quickly. All of these

guidelines on how to create an effective online system design will help the user to use it more and recommend it to other people.

The Americans with Disabilities Act has guidelines on how to create an online system that is accessible to all people. Some barriers to accessibility are poor color contrast, use of only color to provide information, no captions on images or videos, and inaccessible online forms (U.S. Department of Justice Civil Rights Division). Poor color contrast can cause people with limited vision to not be able to read certain text.

Using only color to provide information can cause someone who cannot see color well to not see that a box is red, which might mean that they have to input information into it in order for the system to work correctly.

Color is also important with hyperlinks. Links that have been clicked should have a different color. For example, an unclicked link should be a bright blue color, but a clicked link should be a faded purple color. The faded color looks more “worn out” (Kalbach). This shows the user that they already clicked on that link.

Captions on images and videos will help people to understand what the image or video is about. It shows the purpose of an image and it will help people who cannot hear videos to be able to see the captions of what is being said. It is important to use

font size to emphasize headings and the body. For example, the headings are larger than the body. This increases readability and the user can find what they want to click on easily. It is not recommended to use underlining as emphasis since on online systems, underlines usually mean that the text is a hyperlink. When different font characteristics are used for differentiation, the speed the page is read at decreases by about twenty percent (Leavitt and Shneiderman). In a study of the importance of certain readability factors across age groups of older people, teenagers, and children, it shows that there are factors that are important for all groups. These factors are white spaces, a normal font style, a large font size, and clear headings (Hussain, et al). Line spacing makes sure that lines do not overlap and are able to be read easily.

Making online forms accessible is important. It means that fields should be labeled clearly and have clear instructions. It is also recommended to use text labels instead of just using icons to display the meaning of buttons (Tidwell). A button labeled “Submit” should be clear to the user and should be located at the bottom of the flow of events. For example, if the user fills in a form, there should be a “Submit” button at the bottom of the form. There should also be clear error messages and instructions on

how to fix the error. For example, if the user does not enter information into a required field, they should be instructed to enter information in that field.

Form Input Validation

Form input validation is an important factor in validating information entered by the user. When the user is given a form to input data into, the data should be checked to validate that the data they entered is the correct type (Levis, et. al.). For example, the user should enter text instead of numbers when they are expected to enter an artist's name. When the user is expected to enter information into a field, it is important to check that they entered information there.

People can make errors or typos when entering data into a form (Hayes and Offutt).

When a form has validation checks, it makes sure that the user is entering the correct data. For example, if the user forgets the "@" symbol in their email, the system tells them that what they entered is not a valid email. This will prevent incorrect information from being entered. Also, the fields that the user is filling out should be clear so that they enter the intended information. For example, buttons should be clearly labeled so the user does not select the incorrect button.

Metadata in Online Music Systems

Metadata is data about data. For example, in music streaming services, metadata includes things such as artists names, their songs, what playlists the songs are on, and more (Morris). This is helpful for organizing the data on online systems. It makes sure that the data is located in the correct place so that it can be searched for and used easily. Music streaming services might include metadata such as the performance information (the artist or artists who performed the song), the song lyrics, a descriptor (the title of the song), event information (when and where the song was played) and more (Corthaut, et al). This helps to organize the data so it can be used effectively in the system.

Missing Elements in Existing Online Systems

Although there are some online systems that already exist that are similar to the system that was created, they are missing key elements. For example, there is a website called setlist.fm that has the setlists from tours. However, it only includes each artist's personal setlist. It does not show the openers and the headliners on one playlist. This is inconvenient to the user because they will have to listen to each playlist individually in order to hear the full setlist of the tour. Also, the website does not show the lyrics of the songs. This is proven to be an important factor in listening

to music, and it improves the user's experience. For example, although Spotify has some tour playlists, they do not all contain the lyrics. They are also not highly accessible, since the user will have to search for the name of the tour they are looking for and find the correct playlist among all of the other playlists that will show up. For example, if they search for a certain artist, it is possible that a playlist of their tour does not exist. This will cause them to waste time looking through the results that show up when they search for that artist.

How My Online System Meets These Needs

The created system is an online catalog of music artists, concert setlists, opening and headlining acts, and related information about the songs in concerts. This system is only for setlists and it is easy to see if the artist they search for has a playlist available to listen to in the system. The system also has an option to view the lyrics of the song. This helps the user to have a better listening experience because they are able to understand the songs better. The system also includes the openers and headliner of the tour in one playlist so users can listen to the whole concert in one place. This makes the experience convenient so they do not have to look in more than one place for the

opener's setlists and the headliner's setlist. Since playlists and concerts are both important, the created system combines both of these.

It is shown that user retention on a music streaming service is largely based on the amount of diversity and variety of music in the system (Anderson, et al). This means that having a lot of playlists and songs available on the system is important for people to keep using the system. The system uses features of previously successful music streaming services, such as Spotify. For example, Spotify shows many options to the user of playlists they can choose. The user can click on previously made playlists, or they can create their own. Also, Spotify presents the information to the user in a user-friendly way. The created system's user interface is intuitive. The system allows users to view each other's playlists which will create a community of listeners.

Methodology

The subject that was researched was "An online catalog of music artists concert setlists, opening and headlining acts, and related information about the songs in concerts". The methodology of how this research question was investigated was to build a system that follows the best practices of software design and displays playlists

of concert setlists and shows information about the songs and artists. An iOS application was created to fulfill this objective.

The database used was the MongoDB database. The database in MongoDB has four collections, which are named “artist”, “setlist”, “song”, and “users”. The “artist” collection has fields “_id”, “artistName”, and “artistImage”. The field “_id” has type ObjectId, the field “artistName” has type String, and the field “artistImage” has type String.

The “setlist” collection has fields “_id”, “date_location”, “setlistName”, “setlistImage”, “createdBy”, “tourName”, “headlinerName”, “openerNames”, and “headlinerSongIDs”. The field “_id” has type ObjectId. The field “date_location” is an array of objects, with each containing a date (which has type Date) and a city (which has type String). The field “setlistName” has type String. The field “setlistImage” has type String. The field “createdBy” has type ObjectId, which is a reference to the user who created the setlist. The field “tourName” has type String. The field “headlinerName” has type String. The field “openerNames” is an array of objects, with each containing a “name” (which has type String) and “songs” (an array of ObjectId references to the songs performed by the opener). The field

“headlinerSongIDs” is an array of ObjectId references to the songs performed by the headliner.

The “song” collection has fields “_id”, “songTitle”, “artistID”s, “audio”, and “lyrics”. The field “_id” has type ObjectId. The field “songTitle” has type String. The field “artistIDs” is an array of ObjectId references, which references the artist(s) who performed the song. The field “audio” has type String, which is the link to the audio file of the song. The field “lyrics” has type String, which is the link to the lyrics of the song.

The "users" collection has fields “_id”, “email”, “name”, “password”, and “favoriteSetlists”. The field “_id” has type ObjectId. The field “email” has type String. The field “name” has type String. The field “password” has type String. The field “favoriteSetlists” is an array of ObjectId references to the setlists that the user has marked as favorites.

Node.js and Express were used as the backend of the system. Node.js is a runtime environment that allows JavaScript code outside of the browser. Express is a framework that is built on top of Node.js. It allows you to define routes, send responses, and more. The backend uses Mongoose to connect to the MongoDB

database. The purpose is to define API routes by making requests such as GET or POST. Node.js and Express connect to MongoDB database, handle logic such as formatting, and send back JSON data to Xcode.

There is logic to authenticate input information, such as login information. For example, the system makes sure that the user entered an email in the email field and did not leave it blank. If they do not enter an email, it shows an alert message to the user instructing them to enter an email.

The system has an organized design and presents the data in a user-friendly way. It ensures that the font is large enough for people to see, and that the colors are used in a way that allows the user to easily read the text on the page. For example, there is no dark gray text over a black background, since this is difficult to see due to a lack of contrast.

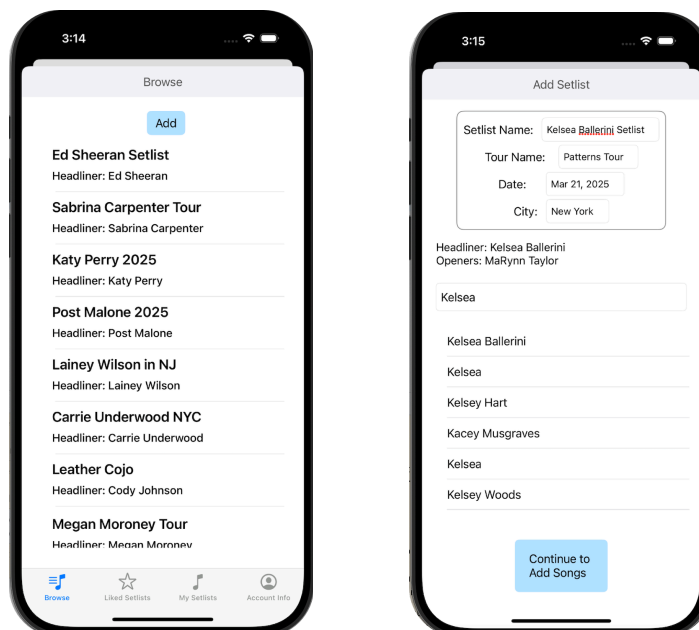
In order to test the online system (before it can possibly be deployed for the public), a local server was used. This allowed the system to be tested before it is available for other people to use. This allows any bugs to be fixed and it allows the thorough development of the system before people can use it. When the system is developed, it can be deployed for public use.

The first screen of the app is a login screen with the name of the app, the logo, an email text field, a password text field, and a login button. It is where the user enters their email and password. There is also an option to register a new user. On the register screen, the user enters their name, email, creates a password, and clicks the “register” button. Then, the user is brought back to the login page. Once the user is logged in, there are four tab options. There is a “Browse” tab, a “Favorites” tab, a “My Setlists” tab, and an “Account Info” tab.

The “Browse” tab is what the user sees when they first log in. They see a list of all of the setlists that have been posted. When the user selects a setlist name from the table, they see the detail view for that setlist. It displays the setlist name, tour name, date of the concert, artist(s) in the setlist, the user’s name who created the setlist, a list of songs in the setlist, and a star icon to select if the user wants to mark it as a favorite. For each song in the setlist, the table shows the song name, a button for the audio link, and a button for the lyrics link.

Also on the “Browse” tab, there is an option for the user to add a concert setlist playlist by clicking a “+” button in the top right of the screen. If the user selects that button, in order to add a concert setlist playlist, the user will enter the setlist name,

name of the tour, date, city, opening artist's name(s) if applicable, and the headliner's name. The application uses Spotify's API to search for artists. The user has the option to add as many opening artists as they want, but at least the headliner is required. They can then click "Continue to add songs" which takes them to the page to add songs in each artist's setlist. On the "add songs" page, the user will see the names of the artists they selected. The headliner has "(Headliner)" next to their name, and the openers have "(Opener)" next to their name. Next to each artist, there is an "Add Song" button. When the user clicks that button, a search bar shows up and they search Spotify to add a song for that artist. The selected song shows up in that artist's table. The user can continue adding songs until they are done, and then they click the "Submit" button to add the setlist to the database. The user is redirected back to the "Browse" tab. Below are images of the "Browse" tab, and Spotify results showing up when adding artists.



The “Favorites” tab has a list of the setlists the user selected as a favorite. The user can select a setlist to see the detail view. The user can unclick the star icon to remove the setlist from their favorites.

The “My Setlists” tab is a screen that shows all of the setlists that the user has created. It is a table of titles of concert setlist playlists that the user has created. The user can select a setlist to see the detail view. The “Account Info” tab displays the logged in users information. It displays the users’ name, email, and has a logout button.

These methods and resources met the needs of the research question and capstone because the methods used the best practices of software design. There are organized and functional app screens. MongoDB was the database used to organize the data. The programming language Swift was used to create the application in Xcode, and Node.js and Express were used to create the backend Node Express files. Postman was used to test the app to make sure it would be functional by testing the API calls. The design principles researched were used together in order to make the system functional and user-friendly. For example, the screens are user-friendly and easy to use. The design of the screens makes the system look interesting by displaying

colors and styles, and it uses the principles set by the Americans with Disabilities Act in order to make the website accessible to everyone. For example, the color of text is contrasted to the color of the background. Also, the font is big enough for people to read easily. These methods and resources answered the research question by allowing the creation of an iOS application that follows the best practices of software design and display playlists of concert setlists.

Reflection Essay

Why This Project is Important

Music is an important part of people's lives, and live concerts provide a specific type of connection between artists and their fans. However, there was previously no mobile application that allowed fans to catalog, share, and relive their favorite concert setlists, particularly with the ability to view openers' and headliners' setlists in one central setlist. Combining these features with audio and lyric links for the user to use, the gap was bridged through the creation of the iOS application "Setify".

The Problem This Project Solved

This research paper looked at how to design a mobile application that allows users to create, browse, and interact with concert setlists in a personalized and accessible way. Although playlist applications exist, they are scattered in content and do not have a focused goal. Also, there was no easy way for users to organize songs by both headliners

and openers in one setlist, and store that setlist in a collection of other setlists just like itself. “Setify” solves this problem by allowing users to catalog entire concert setlists and access essential song data such as audio and lyrics, all within one central platform.

How the Project Unfolded Over Time

In Fall 2023, I took the class HO 296 where I conceptualized and researched the idea. I researched existing similar applications and analyzed what was missing from them. I reviewed existing literature on topics of user interface design, best practices of software development, and more. I developed the idea of creating a setlist application that allows openers and headliners to be in the same setlist, includes audio and lyrics information, and allows users to browse through the posted setlists.

In Spring 2024, I took the class HO 396 where I created the data models for the application. I designed the MongoDB database schema and developed a wireframe for the application screens. I also researched how to create the backend Node.js Express API, and how to include the use of Spotify’s API.

In Spring 2025, I took the class HO 496 where I designed and implemented the application. I built the Node.js Express backend, the iOS frontend using Xcode and Swift, and integrated Spotify’s API into the application. I also tested the application and ensured a user-friendly experience in the application.

How Ideas were Generated

Ideas were generated through my passion for music, user-centered design, and feedback from peers and mentors. I made design decisions based on the research conducted and user expectations. For example, I chose to include the ability to have audio and lyrics

links for songs so that users would be able to relive the concert experience through interaction. I focused on a clean user interface with intuitive navigation.

Actions, Meeting, and Significant Roadblocks

Throughout the project, I had regular meetings with my capstone advisor. Significant actions that were taken include designing the backend API, building custom UI views to allow users to create setlists with multiple artists, using Spotify's API, and implementing them together to produce the functioning application. Significant roadblocks included designing the system to allow multiple artists to be added to a setlist without complicating the user interface. I solved this through separating the artists into sections in a tableView, which made the user interface cleaner. Another roadblock was ensuring data consistency between Xcode and the backend API. I had to make sure that I represented the data types correctly. These roadblocks were overcome with researching the issue, conducting testing, debugging the code, and getting input from my capstone advisor.

Overall Themes, Plans, and Outcomes

The overall themes of the capstone project were creating a user-friendly application, using the best practices of software design, and the benefits of creating a system like this application. Previously, there was no application to store concert setlist playlists with opener and headliner songs and related song information. This capstone project solved that problem.

Noteworthy Moments

Getting Spotify's API to integrate with the system was an important step. Seeing artists from Spotify displayed in the application was an important part in moving forward in the

process. Also, testing the application once it was completed felt rewarding. Setify solves the problem that other applications have by being an online catalog of music artists' concert setlists, having a way to view lyrics and audio of songs, and having opening and headlining acts.

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